

Drug Utilization Review (DUR) in Elderly Patients: Medicare Part D Real-World Evidence Analysis

Problem Statement

Older adults (≥ 65 years) are disproportionately exposed to medication-related harm due to polypharmacy, age-related physiological changes, multimorbidity, and prolonged medication use. High-risk medications including benzodiazepines, opioids, anticoagulants, nonsteroidal anti-inflammatory drugs (NSAIDs), anticholinergics, and hypoglycemic agents are associated with preventable adverse outcomes such as falls, bleeding, respiratory depression, delirium, renal injury, and hospitalization.

Despite established prescribing safety frameworks such as the Beers Criteria and STOPP/START principles, potentially inappropriate medications continue to be widely prescribed in elderly populations. The growing burden of medication-related harm creates substantial clinical and economic challenges for healthcare systems, particularly in populations with complex chronic disease management needs.

A major challenge in geriatric pharmacotherapy is identifying:

- Which high-risk medications contribute most to prescribing burden?
- Which medication classes generate the highest healthcare costs?
- Which provider specialties are associated with elevated high-risk prescribing?
- What preventable adverse outcomes are most likely occurring due to medication exposure?
- Which medications should be prioritized for deprescribing interventions?

Addressing these questions is essential to improving medication safety, reducing avoidable adverse drug events, optimizing prescribing practices, and minimizing unnecessary healthcare expenditure among older adults.

Project Objective

This project conducts a **Drug Utilization Review (DUR)** using **real-world Medicare Part D prescribing data** to evaluate high-risk medication exposure among elderly beneficiaries (≥ 65 years).

The analysis applies **Beers Criteria** and **STOPP/START-informed medication safety principles** to identify potentially inappropriate medications and quantify their utilization burden, financial impact, preventable risks, and deprescribing opportunities.

The primary objective is to generate evidence that supports:

- Safer prescribing practices in elderly populations
 - Pharmacist-led medication review programs
 - Targeted deprescribing interventions
 - Provider specialty risk identification
 - Prevention of medication-related adverse outcomes
 - Healthcare cost optimization
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Dataset

Source: CMS Medicare Part D Prescriber Public Use File

The dataset contains prescribing-level information on medications prescribed to Medicare beneficiaries, including:

- Provider specialty
- Drug generic name
- Number of claims
- Drug spending
- Elderly beneficiary utilization (≥ 65 years)

The analysis specifically focuses on elderly utilization metrics using:

- `GE65_Tot_Clms`
- `GE65_Tot_Drug_Cst`
- `GE65_Tot_Benes`

to evaluate prescribing burden among older adults.

Methodology

The project was conducted in multiple analytical stages using **SQL**, **real-world evidence (RWE) methods**, and **Tableau dashboards**.

Step 1: Elderly Drug Utilization Extraction

Prescribing data for elderly beneficiaries (≥ 65 years) was extracted from the Medicare Part D dataset to create an elderly-focused utilization table.

Step 2: High-Risk Medication Identification

A curated reference table of **37 high-risk medications** was developed using:

- Beers Criteria
- STOPP/START principles
- Clinical deprescribing literature
- Geriatric medication safety evidence

Each medication was categorized according to:

- Drug class
- Risk level
- High-risk reason
- Likely adverse outcome
- Primary risk category
- Secondary risk category
- Deprescribing recommendations

Step 3: Drug Utilization Review (DUR) Analysis

High-risk medications were merged with elderly prescribing data to create a structured DUR dataset for downstream analysis.

Step 4: Analytical Framework

Seven major analyses were conducted:

1. **High-Risk Medication Burden**
Quantified prescribing burden by risk level.
2. **Drug Class Burden Analysis**
Identified medication classes contributing most to utilization and cost burden.
3. **Top High-Risk Drugs**
Ranked medications by claims and economic burden.
4. **Cost Burden Analysis**
Evaluated high-cost medications and cost-per-claim.
5. **Provider Specialty Risk Analysis**
Assessed specialty-specific prescribing burden and medication-related risk.
6. **Preventable Adverse Outcomes**
Identified likely medication-related harms associated with prescribing patterns.
7. **Deprescribing Opportunity Analysis**
Prioritized medications suitable for pharmacist-led medication review and deprescribing.

Key Research Questions

This project aimed to answer the following questions:

1. Which high-risk medications are most frequently prescribed in elderly patients?
 2. Which drug classes generate the highest utilization and healthcare spending?
 3. Which provider specialties contribute most to high-risk medication exposure?
 4. What preventable adverse outcomes are most strongly associated with elderly prescribing patterns?
 5. Which medications should be prioritized for deprescribing interventions?
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Tools & Technologies

- **SQL (SQLite)**; Data cleaning, transformation, and analytical querying
 - **Tableau**; Dashboard creation and visualization
 - **Real-World Evidence (RWE) Framework**
 - **Drug Utilization Review (DUR) Methodology**
 - **Beers Criteria**
 - **STOPP/START-informed prescribing principles**
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Expected Clinical Impact

The findings of this project may support:

- Reduction of preventable adverse drug events
 - Improved medication safety in older adults
 - Pharmacist-led deprescribing interventions
 - Safer prescribing policies for elderly populations
 - Reduced healthcare expenditure from medication-related harm
 - Better risk stratification for provider specialties
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Project Outcome

This project demonstrated substantial high-risk medication exposure among elderly Medicare beneficiaries and identified multiple opportunities for medication optimization.

The findings highlight the importance of targeted medication review, deprescribing strategies, and specialty-specific prescribing interventions to reduce preventable harm and improve outcomes in geriatric populations.